Zhengni Hu

Postdoc, School of Mathematical Sciences
Shanghai Jiao Tong University

Email: zhengni_hu@sjtu.edu.cn

Personal Information

- Date of Birth Oct. 01, 1996.
- Place of Birth Changsha, Hunan, China.
- Citizenship China.

Research Interests

Nonlinear Partial Differential Equations and Geometric Analysis

- Nonlinear partial differential equations arising in geometry and mathematical physics, including mean field equations, Liouville-type systems, super-Liouville equations, and related models
- Qualitative and geometric analysis of solutions to nonlinear PDEs, with a focus on singular phenomena such as blow-up and concentration
- Variational methods, finite-dimensional reduction and "critical point at infinity" techniques for nonlinear PDEs

Work Experience

Wen-Tsun Wu Assistant Professorship (Postdoctoral Fellow)
Shanghai Jiao Tong University Dec. 2024 – present
Advisors: Prof. Miaomiao Zhu & Prof. Tongrui Wang

Education

- **2021–2024** Ph.D. in Mathematics, Justus Liebig University Giessen, Germany.
  - Dissertation: *Blow-up Solutions for Mean Field Equations and Toda Systems on Riemann Surfaces*
  - Advisors: Prof. Thomas Bartsch & Prof. Mohameden Ahmedou
- **2018–2021** M.Sc. in Mathematics, Peking University, Beijing, China.
  - Thesis: *On Elliptic Solutions of a Generalized Monge–Ampère Equation*
  - Supervisor: Prof. Zhifei Zhang
- **2014–2018** B.Sc. in Mathematics and Applied Mathematics, Beijing Normal University, Beijing, China.
  - Thesis: *Compact Embedding Between Besov Spaces and Triebel–Lizorkin Spaces on Metric Spaces* (evaluated as an excellent bachelor thesis)
  - Supervisor: Prof. Wen Yuan

Publications

1. Blow-up solutions for the steady state of the Keller–Segel system on Riemann surfaces.
   M. Ahmedou, T. Bartsch, and Z. Hu, *Calc. Var. Partial Differential Equations*, 2026.
   DOI: 10.1007/s00526-026-03313-5
2. A degree-counting formula for a Keller–Segel equation on a surface with boundary.
   M. Ahmedou, Z. Hu and H. Wang. *Commun. Contemp. Math.*, 2026.
   DOI:10.1142/S0219199726500392

3. Blow-up solutions for general Toda systems on Riemann surfaces.
   Z. Hu and M. Zhu, *J. Differential Equations*, 2026. DOI: 10.1016/j.jde.2026.114145
4. The Morse property of limit functions appearing in mean field equations on surfaces with boundary.
   Z. Hu and T. Bartsch, *J. Geom. Anal.*, 2024. DOI: 10.1007/s12220-024-01664-z

Preprints

1. Blow-up solutions for mean field equations with Neumann boundary conditions on Riemann surfaces. with T. Bartsch and M. Ahmedou. arXiv:2408.16917 v2 .
2. Partial blow-up phenomena in the SU(3) Toda system on Riemann surfaces. with T. Bartsch and M. Ahmedou. arXiv:2408.17372.
3. Blow-up solutions for mean field equations with non-quantized singularities on Riemann surfaces with boundary. with M. Ahmedou and M. Zhu. arXiv:2602.04790.

Teaching

Mathematical Methods in Physics — Fall 2025 – Spring 2026
Lecturer, Shanghai Jiao Tong University

Talks

- Seminar on PDEs, Chongqing Jiao Tong University, China
  May 2026
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
  Feb. 2026
- Seminar on Analysis and Geometry, Shanghai Jiao Tong University, China
  Mar. 2025
- The 1st International Young Scholars Forum, Yunnan Normal University, China
  Jan. 2025
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
  Jul. 2024
- Seminar on Analysis and Geometry, Justus Liebig University Giessen, Germany
  Jun. 2024
- VIII Symposium on Nonlinear Analysis, Nicolaus Copernicus University, Poland
  Jun. 2024
- Summer School on Calculus of Variations: Variational and Geometrical Methods, University of Warsaw, Poland
  Jun. 2024
- Atlantic Conference on Nonlinear Partial Differential Equations, Instituto Superior Técnico, Portugal (Poster Presentation)
  Nov. 2023

Grants and Fellowships

- *China Scholarship Council (CSC) Fellowship* — 2021–2024
  No. 202106010046.

Last updated: June 10, 2026